

```
1 #include <iostream>
2 #include <fstream>
3 #include <vector>
4 #include <sstream>
5 #include <iomanip>
6
7 using namespace std;
8
9 // Define the classes
10
11 class Name {
12 public: //SC1
13     Name(const string& first, const string& last) : firstName(first),
14         lastName(last) {}
15     string getFullName() const {
16         return firstName + " " + lastName; //SC2
17     }
18 private: //SC3
19     string firstName;
20     string lastName;
21 };
22
23 class CarInfo {
24 public: //SC4
25     CarInfo(const string& make, const string& model, const string& plate)
26         : carMake(make), carModel(model), licensePlate(plate) {}
27     string getInfo() const {
28         return carMake + " " + carModel + " " + licensePlate;
29     }
30 private: //SC6
31     string carMake;
32     string carModel;
33     string licensePlate;
34 };
35
36 class Payment {
37 public: //SC7
38     Payment(double ticketPayment, const string& creditCard) :
39         ticketPayment(ticketPayment), creditCardNumber(creditCard) {}
40     string getPaymentInfo() const {
41         stringstream ss; //8
42         ss << "$" << fixed << setprecision(2) << ticketPayment << " " <<
43             creditCardNumber;
44         return ss.str();
45     }
46 private: //SC9
47     double ticketPayment;
48     string creditCardNumber;
49 };
50
51 int main() {
52     // Create a Name object
53     Name n("John", "Doe");
54     cout << n.getFullName() << endl;
55
56     // Create a CarInfo object
57     CarInfo c("Ford", "Mustang", "ABC123");
58     cout << c.getInfo() << endl;
59
60     // Create a Payment object
61     Payment p(10.50, "1234567890123456");
62     cout << p.getPaymentInfo() << endl;
63
64     return 0;
65 }
```

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47
48 class Customer {
49 public: //SC10
50     Customer(const Name& n, const CarInfo& c, const Payment& p) : name(n), ↗
        carInfo(c), payment(p) {}
51     void printInfo() const { //SC11
52         cout << "Name: " << name.getFullName() << endl;
53         cout << "Car Information: " << carInfo.getInfo() << endl;
54         cout << "Speeding Ticket Payment: " << payment.getPaymentInfo() << ↗
            endl;
55         cout << endl;
56     }
57 private: //SC12
58     Name name;
59     CarInfo carInfo;
60     Payment payment;
61 };
62
63 // Driver class
64 class TicketDriver {
65 public:
66     void run() { //SC13
67         readDataFromFile();
68         printRecords();
69     }
70
71 private:
72     void readDataFromFile() { //SC14
73         ifstream file("Names.txt");
74         if (!file.is_open()) {
75             cerr << "Error opening file." << endl;
76             return;
77         }
78
79         string line;
80         while (getline(file, line)) {
81             stringstream ss(line);
82             string firstName, lastName, carMake, carModel, licensePlate, ↗
                creditCardNumber;
83             double ticketPayment;
84
85             if (getline(ss, firstName, ',') && getline(ss, lastName, ',') ↗
                && //SC15
86                 getline(ss, carMake, ',') && getline(ss, carModel, ',') &&
87                 getline(ss, licensePlate, ',') && ss >> ticketPayment && ↗
                    ss.ignore() &&
88                 getline(ss, creditCardNumber)) {
89                 Name name(firstName, lastName);
90                 CarInfo carInfo(carMake, carModel, licensePlate);
```

```
91         Payment payment(ticketPayment, creditCardNumber);
92         customers.emplace_back(name, carInfo, payment);
93     }
94     else {
95         cerr << "Error parsing line: " << line << endl; //SC16
96     }
97 }
98
99 file.close(); //SC17
100
101 void printRecords() const {
102     int lineNumber = 1;
103     cout << "*****" << endl;
104     cout << "**   BPA Speeding Ticket Database   **" << endl;
105     cout << "*****" << endl;
106     cout << "*****" << endl;
107     cout << "                               " << endl;
108
109     for (const auto& customer : customers) {
110         cout << "*****>>>   " << setw(2) << lineNumber << "          ↗
111             <<<*****" << endl; //SC19
112         customer.printInfo(); //SC18
113         lineNumber++;
114     }
115     cout << "*****" << endl;
116 }
117
118 vector<Customer> customers;
119 };
120
121 int main() {
122     TicketDriver driver;
123     driver.run();
124     return 0;
125 }
```